

1. Administrative & Study Identification

Full Study Title: Single-arm, Open Label, Phase II Study of MBG453 (Sabatolimab) Added to FDA Approved Hypomethylating Agents of Investigator's Choice (IV/SC/Oral) for Patients With Intermediate, High or Very High Risk Myelodysplastic Syndrome (MDS) as Per IPSS-R Criteria (US Multi-center) **Short Title/ Acronym:** STIMULUS MDS-US : Sabatolimab Added to HMA in Higher Risk MDS **Protocol Number:** CMBG453B1US01 **NCT Number:** NCT04878432 **Posting ID:** 409505120389313776 **Sponsor/Company Name:** Novartis / Novartis Pharmaceuticals Corporation **Investigational Product:** Sabatolimab (MBG453) in combination with Hypomethylating Agents (HMAs: Azacitidine [Trade names: Vidaza, Onureg; ATC: L01BC07] or Decitabine) **Phase of Development:** Phase II **Study Status:** TERMINATED **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety profile of MBG453 (sabatolimab) in combination with FDA-approved HMAs of the investigator's choice (IV Decitabine or Azacitidine / SC Azacitidine / Oral Decitabine (cedazuridine combination [INQOVI])) in patients with higher-risk MDS. **Secondary Objectives:** To evaluate the complete remission (CR) rate, progression-free survival (PFS) (including death and disease progression, and 50th percentile).

2.2 Background & Rationale Patients with intermediate-, high-, or very-high-risk myelodysplastic syndrome (I/H/vHR-MDS) who are unfit for hematopoietic stem cell transplant (HSCT) experience poor outcomes with standard hypomethylating agents (HMAs), with many progressing within 2 years. Sabatolimab is a novel immuno-myeloid therapy that binds to TIM-3 on immune cells and leukemic stem cells (LSCs), facilitating immune activation and potentially impeding LSC self-renewal. New combination therapies are urgently needed to provide durable responses and survival benefits with favorable safety profiles.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Non-randomized)

3.2 Planned Enrollment & Duration Target \$N\$: 39 participants
Number of Sites: US Multi-center **Study Status & Termination**

Details: The study was terminated early. Recruitment was halted on 30-Sep-2022 based on results from the Phase II STIMULUS MDS-1 study (not due to safety findings or concerns). On 11-Jan-2024, Novartis notified investigators that the sabatolimab development program was terminated based on a Dec-2023 decision. The actual last patient last visit date was 1-Sep-2024.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the date of signing the informed consent form (ICF). 2. Morphologically confirmed diagnosis of primary or secondary Myelodysplastic Syndrome (MDS) based on the 2016 WHO classification. 3. International Prognostic Scoring System-Revised (IPSS-R) risk category of Intermediate (> 3 to ≤ 4.5 points), High (> 4.5 to ≤ 6 points), or Very High (> 6 points). 4. Not suitable at the time of Screening for immediate myeloablative/chemotherapy or HSCT based on Investigator assessment. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1 or 2. 6. AST and ALT $\leq 3 \times$ upper limit of normal (ULN); Total bilirubin $\leq 2 \times$ ULN (except isolated Gilbert syndrome); Estimated glomerular filtration rate ≥ 30 mL/min/1.73 m².

4.2 Exclusion Criteria 1. Prior exposure to TIM-3 directed therapy at any time. (Prior immune checkpoint inhibitors allowed if last dose was > 4 months prior). 2. Previous treatment for higher-risk MDS with chemotherapy or other antineoplastic agents including lenalidomide and HMAs (up to 1 cycle of HMAs was permitted; previous hydroxyurea permitted). 3. Diagnosis of acute myeloid leukemia (AML), chronic myelomonocytic leukemia (CMML), or primary or secondary myelofibrosis based on 2016 WHO classification. 4. History of organ transplant or allogeneic HSCT. 5. Prior malignancy, with exceptions for lower-risk MDS (treated supportively or failed lenalidomide) or adequately treated malignancies requiring no ongoing systemic anticancer therapy. 6. Active, severe, or uncontrolled infections or severe concurrent medical conditions.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sabatolimab (MBG453) IV administered in combination with an FDA-approved HMA of the investigator's choice (Azacitidine IV/SC, Decitabine IV, or Oral Decitabine/Cedazuridine [INQOVI]). **Route of Administration:** Intravenous (IV) for sabatolimab; IV, Subcutaneous (SC), or Oral for the HMA partner. **Duration of Treatment:** 1. Core phase for up to 12 months. 2. Extension phase for efficacy and/or survival status (up to 12 months after the core phase).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for MDS (e.g., anti-infectives, transfusions). **Prohibited:** Other concurrent investigational agents, additional systemic anti-cancer therapies, or immunotherapies outside the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, IPSS-R Scoring, Baseline Bone Marrow/Labs
Baseline	Day 1	Enrollment, First Dose of Sabatolimab + HMA
Interim	Q4W (Cycles)	Safety Labs, Efficacy Assessment (Bone marrow aspirates/biopsies as clinically indicated), AE monitoring
End of Core Phase	Month 12	Evaluation for Extension Phase, Survival/Response follow-up
Follow-up	Post-Tx	Safety follow-up for 30 days following the last dose of azacitidine, decitabine, or INQOVI, or 150 days following the last dose of sabatolimab, whichever is later.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number of Participants With Treatment Emergent Adverse Events and Serious Adverse Events. **Measurement Method:** Clinical observation, vital signs, physical examinations, and laboratory assessments evaluated continuously during the study to monitor safety profile.

7.2 Secondary Endpoints * Number of Participants With Complete Remission According to International Working Group (IWG) for MDS (2006) With MBG453 (Sabatolimab) in Combination With HMAs (IV/SC/Oral) by 12 Months. * Number of Participants With Progression Free Survival - Death and Disease Progression. * Progression Free Survival - 50th Percentile.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs and SAEs from the time of informed consent through the designated post-treatment follow-up period. **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External, independent Data Monitoring Committee to evaluate ongoing safety signals and benefit-risk profiles.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose of study medication) and Full Analysis Set (FAS) for efficacy. **Sample Size Justification:** A target enrollment of \$N=39\$ was planned to provide descriptive power to evaluate the safety profile and provide initial estimates of complete remission and PFS in this specific US cohort prior to early termination. **Handling of Missing Data:** Efficacy endpoints like PFS and OS analyzed using standard methods, with patients lost to follow-up censored at the date of last contact.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine site visits (remote and onsite) by the Sponsor's clinical research associates (CRAs) to ensure protocol adherence and data accuracy. **Audit Trail:** Electronic Case Report Forms (eCRFs) utilized with built-in audit trails, data clarification forms, and scheduled database locks.

11. Study Identifiers & Governance

Primary ID: {{CMBG453B1US01}} **Posting ID:** {{409505120389313776}}
Registry Number: {{NCT04878432}} **Sponsor/Lead Organization:** {{Novartis / Novartis Pharmaceuticals Corporation}} **Study Title:** {{STIMULUS MDS-US : Sabatolimab Added to HMA in Higher Risk MDS}}
Official Regulatory Title: {{Single-arm, Open Label, Phase II Study of MBG453 (Sabatolimab) Added to FDA Approved Hypomethylating Agents of Investigator's Choice (IV/SC/Oral) for Patients With Intermediate, High or Very High Risk Myelodysplastic Syndrome (MDS) as Per IPSS-R Criteria (US Multi-center) (STIMULUS MDS-US)}}

12. Clinical Framework (Myelodysplastic Syndromes Specific)

Primary Condition: {{Myelodysplastic Syndrome (MDS)}} **Medical Classifications:** {{MeSH: D009190 | HPO: HP:0002863 | SNOMED: 109995007}} **Diagnosis Threshold:** {{IPSS-R score > 3 points (Intermediate, High, or Very High)}} **Medical Methodology:** {{Immuno-myeloid targeted therapy combined with Hypomethylating Agents}} **Optimization Target:** {{Induction of Complete Remission (CR) according to IWG (2006) and Progression Free Survival}}

13. Study Procedures & Methodology

Management Pathway: {{Combination therapy of Sabatolimab + HMA}}
Education Protocol (HCP): {{Protocol training and AE management guidelines for novel immunotherapies}} **Education Protocol (Patient):** {{Informed consent regarding risks of investigational immunotherapy and standard HMA side effects}} **Quality Audit Mechanism:** {{Routine Sponsor CRA monitoring and independent data review}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 12 months (Core) + 12 months (Extension) + Post-treatment safety follow-up (30 days post-HMA or 150 days post-sabatolimab, whichever is later)}} **Onsite Visit Cadence:** {{Day 1 of each treatment cycle and as clinically indicated for HMAs}} **Remote Monitoring Frequency:** {{Standard interval monitoring per Sponsor clinical operations plan}} **Checklist Review Interval:** {{Cycle-by-cycle response and toxicity evaluations}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				